I U S

Faculté des Sciences et des Technologies

Administration des Systèmes Informatiques I

TD n° 6

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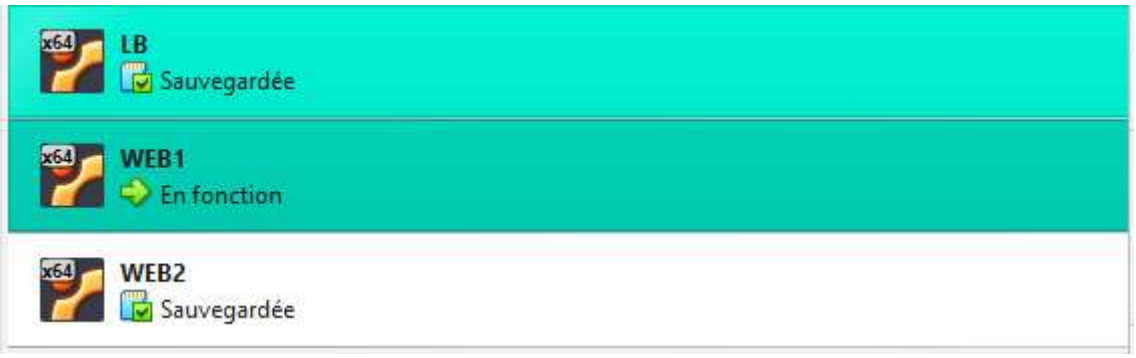
Serveur Web Haute Disponibilité

1. Déployer une architecture web résiliente
2. Configurer un site dynamique PHP
3. Optimiser les performances (cache)
4. Sécuriser avec HTTPS
5. Tester la montée en charge

E-commerce HA + Paiement simulé

Machines



ÉTAPE 1 — Installer Nginx + PHP sur WEB1

```
web1@web1:~$ sudo apt update
sudo apt install nginx php-fpm -y
[sudo] password for web1:
Atteint :1 http://security.ubuntu.com/ubuntu noble-security InRelease
Atteint :2 http://us.archive.ubuntu.com/ubuntu noble InRelease
Atteint :3 http://us.archive.ubuntu.com/ubuntu noble-updates InRelease
Atteint :4 http://us.archive.ubuntu.com/ubuntu noble-backports InRelease
Lecture des listes de paquets... Fait
Construction de l'arbre des dépendances... Fait
Lecture des informations d'état... Fait
Tous les paquets sont à jour.
Lecture des listes de paquets... Fait
Construction de l'arbre des dépendances... Fait
```

```
web1@web1:~$ sudo systemctl status nginx
● nginx.service - A high performance web server and a reverse proxy server
     Loaded: loaded (/usr/lib/systemd/system/nginx.service; enabled; preset: enabled)
     Active: active (running) since Sun 2026-04-12 21:08:01 UTC; 2min 51s ago
       Docs: man:nginx(8)
   Main PID: 14128 (nginx)
      Tasks: 3 (limit: 4601)
     Memory: 2.4M (peak: 5.3M)
        CPU: 229ms
     CGroup: /system.slice/nginx.service
             ├─14128 "nginx: master process /usr/sbin/nginx -g daemon on; master_process on;"
             ├─14130 "nginx: worker process"
             └─14131 "nginx: worker process"

avril 12 21:08:01 web1 systemd[1]: Starting nginx.service - A high performance web server and a reverse
avril 12 21:08:01 web1 systemd[1]: Started nginx.service - A high performance web server and a reverse p
web1@web1:~$
```

```
web1@web1:~$ sudo systemctl list-unit-files | grep -i php
php8.3-fpm.service                         enabled         enabled
phpsessionclean.service                    static          -
phpsessionclean.timer                      enabled         enabled
web1@web1:~$
```

Création du site e-commerce sur WEB1

```
web1@web1:~$ sudo mkdir -p /var/www/shop
web1@web1:~$ sudo nano /var/www/shop/index.php
web1@web1:~$
```

Permissions

```
web1@web1:~$ sudo chown -R www-data:www-data /var/www/
web1@web1:~$ sudo chmod -R 755 /var/www/
web1@web1:~$
```

Configuration de Nginx

```
web1@web1: ~
web1@web1:~$ sudo nano /etc/nginx/sites-available/shop
[sudo] password for web1:
web1@web1:~$ cat /etc/nginx/sites-available/shop
server {
    listen 80;
    server_name shop.devcloud.local;

    root /var/www/shop;
    index index.php;

    location / {
        try_files $uri $uri/ =404;
    }

    location ~ \.php$ {
        include snippets/fastcgi-php.conf;
        fastcgi-pass unix:/run/php/php8.1-fpm.sock;
    }
}
web1@web1:~$
```

Activation du site

```
web1@web1:~$ sudo ln -s /etc/nginx/sites-available/shop /etc/nginx/sites
-enabled/
web1@web1:~$ sudo rm /etc/nginx/sites-enabled/default
```

```
web1@web1:~$ sudo nginx -t
nginx: the configuration file /etc/nginx/nginx.conf syntax is ok
nginx: configuration file /etc/nginx/nginx.conf test is successful
web1@web1:~$ sudo systemctl restart nginx
web1@web1:~$
```

ÉTAPE 2 — Installer Nginx + PHP sur WEB2

```
web2@web2: ~
web2@web2:~$ sudo apt update
sudo apt install nginx php-fpm -y
[sudo] password for web2:
Atteint :1 http://archive.ubuntu.com/ubuntu noble InRelease
Atteint :2 http://security.ubuntu.com/ubuntu noble-security InRelease
Atteint :3 http://archive.ubuntu.com/ubuntu noble-updates InRelease
Atteint :4 http://archive.ubuntu.com/ubuntu noble-backports InRelease
Réception de :5 http://archive.ubuntu.com/ubuntu noble/main Translation-
fr [491 kB]
Réception de :6 http://archive.ubuntu.com/ubuntu noble/restricted Transl
ation-fr [3 292 B]
Réception de :7 http://archive.ubuntu.com/ubuntu noble/universe Translat
ion-fr [3 898 kB]
Réception de :8 http://archive.ubuntu.com/ubuntu noble/multiverse Transl
ation-fr [89,0 kB]
4 481 ko réceptionnés en 5s (844 ko/s)
Lecture des listes de paquets... Fait
Construction de l'arbre des dépendances... Fait
```

```
web2@web2: ~
web2@web2:~$ systemctl status nginx
● nginx.service - A high performance web server and a reverse proxy ser>
     Loaded: loaded (/usr/lib/systemd/system/nginx.service; enabled; pr>
     Active: active (running) since Sun 2026-04-12 22:06:50 UTC; 36min >
       Docs: man:nginx(8)
   Main PID: 6669 (nginx)
      Tasks: 3 (limit: 4601)
     Memory: 2.4M (peak: 5.3M)
        CPU: 182ms
     CGroup: /system.slice/nginx.service
             ├─6669 "nginx: master process /usr/sbin/nginx -g daemon on>
             ├─6672 "nginx: worker process"
             └─6673 "nginx: worker process"

avril 12 22:06:50 web2 systemd[1]: Starting nginx.service - A high perf>
avril 12 22:06:50 web2 systemd[1]: Started nginx.service - A high perfo>

web2@web2:~$ systemctl list-unit-files | grep -i php
php8.3-fpm.service                         enabled         enabled
phpsessionclean.service                    static          -
phpsessionclean.timer                      enabled         enabled
web2@web2:~$
```

Création du site e-commerce sur WEB2

```
web2@web2:~$ sudo mkdir -p /var/www/shop
[sudo] password for web2:
web2@web2:~$ sudo nano /var/www/shop/index.php
web2@web2:~$
```

Permissions

```
web2@web2:~$ sudo chown -R www-data:www-data /var/www/
web2@web2:~$ sudo chmod -R 755 /var/www/
web2@web2:~$
```

Configuration de Nginx

```
web2@web2:~$ sudo nano /etc/nginx/sites-available/shop
web2@web2:~$ cat /etc/nginx/sites-available/shop
server {
    listen 80;
    server_name shop.devcloud.local;

    root /var/www/shop;
    index index.php;

    location / {
        try_files $uri $uri/ =404;
    }

    location ~ \.php$ {
        include snippets/fastcgi-php.conf;
        fastcgi_pass unix:/run/php/php8.3-fpm.sock;
    }
}
web2@web2:~$
```

Activation du site

```
web2@web2:~$ sudo ln -s /etc/nginx/sites-available/shop /etc/nginx/sites
-enabled/
web2@web2:~$ sudo rm /etc/nginx/sites-enabled/default
web2@web2:~$ sudo nginx -t
nginx: the configuration file /etc/nginx/nginx.conf syntax is ok
nginx: configuration file /etc/nginx/nginx.conf test is successful
web2@web2:~$ sudo systemctl restart nginx
web2@web2:~$
```

ÉTAPE 3 — Configuration de Load Balancer

Ps : J'ai utilisé mon serveur initial comme load balancer

```
PS C:\Users\Christy> ssh -p 2222 christy@192.168.56.113
christy@192.168.56.113's password:
Welcome to Ubuntu 24.04.4 LTS (GNU/Linux 6.8.0-107-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:     https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro

System information as of jeu. 02 avril 2026 05:35:16 UTC

  System load:             0.01
  Usage of /:              43.9% of 11.21GB
  Memory usage:            11%
  Swap usage:              0%
  Processes:               123
  Users logged in:         1
  IPv4 address for enp0s3: 10.0.2.15
  IPv6 address for enp0s3: fd17:625c:f037:2:a00:27ff:febd:76ab
```

Configuration de Nginx

```
christy@christy:~$ sudo systemctl status nginx
[sudo] password for christy:
● nginx.service - A high performance web server and a reverse proxy ser>
     Loaded: loaded (/usr/lib/systemd/system/nginx.service; enabled; pr>
     Active: active (running) since Thu 2026-04-02 05:37:56 UTC; 1 week>
       Docs: man:nginx(8)
    Process: 147723 ExecStartPre=/usr/sbin/nginx -t -q -g daemon on; ma>
    Process: 147730 ExecStart=/usr/sbin/nginx -g daemon on; master_proc>
   Main PID: 147760 (nginx)
      Tasks: 3 (limit: 4601)
     Memory: 2.4M (peak: 5.3M)
        CPU: 171ms
     CGroup: /system.slice/nginx.service
             ├─147760 "nginx: master process /usr/sbin/nginx -g daemon >
             ├─147762 "nginx: worker process"
             └─147763 "nginx: worker process"

avril 02 05:37:56 christy systemd[1]: Starting nginx.service - A high p>
avril 02 05:37:56 christy systemd[1]: Started nginx.service - A high pe>
lines 1-17/17 (END)
christy@christy:~$
```

```
christy@christy:~$ sudo nano /etc/nginx/nginx.conf
christy@christy:~$ cat /etc/nginx/nginx.conf
events {}

http {
    upstream shop_backend {
        ip_hash;
        server 192.168.56.114;
        server 192.168.56.115;
    }

    server {
        listen 80;
        server_name shop.devcloud.local;

        location / {
            proxy_pass http://shop_backend;
            proxy_set_header Host $host;
            proxy_set_header X-Real-IP $remote_addr;
        }
    }
}
christy@christy:~$
```

```
christy@christy:~$ sudo nginx -t
nginx: the configuration file /etc/nginx/nginx.conf syntax is ok
nginx: configuration file /etc/nginx/nginx.conf test is successful
christy@christy:~$ sudo systemctl restart nginx
christy@christy:~$
```

ÉTAPE 4 — Tester depuis Windows

personnes.csv | td3_graphes.txt | hosts

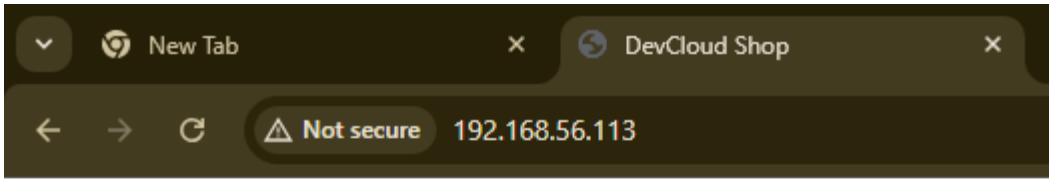
File Edit View H1

```
#
127.0.0.1 localhost
192.168.56.103 cybersecurite.ht
192.168.56.107 centreabel.ht
::1 localhost
192.168.56.113 shop.devcloud.local
```

```
PS C:\Users\Christy> ping 192.168.56.113

Envoi d'une requête 'Ping'  192.168.56.113 avec 32 octets de données :
Réponse de 192.168.56.113 : octets=32 temps=1 ms TTL=64
Réponse de 192.168.56.113 : octets=32 temps=1 ms TTL=64
Réponse de 192.168.56.113 : octets=32 temps=1 ms TTL=64
Réponse de 192.168.56.113 : octets=32 temps=1 ms TTL=64

Statistiques Ping pour 192.168.56.113:
    Paquets : envoyés = 4, reçus = 4, perdus = 0 (perte 0%),
Durée approximative des boucles en millisecondes :
    Minimum = 1ms, Maximum = 1ms, Moyenne = 1ms
PS C:\Users\Christy>
```



🛒 DevCloud Shop

Serveur : web2

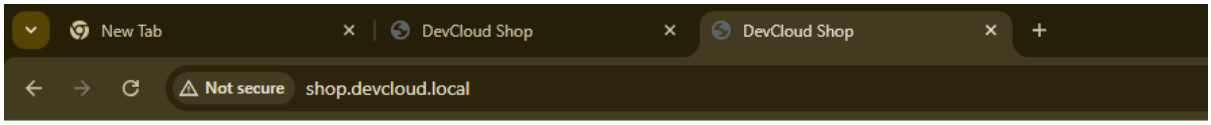
Catalogue Produits

- Laptop - 800 € Ajouter
- Smartphone - 500 € Ajouter
- Headphones - 100 € Ajouter

🛒 Panier

- Panier vide

Total : 0 €



🛒 DevCloud Shop

Serveur : web2

Catalogue Produits

- Laptop - 800 € Ajouter
- Smartphone - 500 € Ajouter
- Headphones - 100 € Ajouter

🛒 Panier

- Panier vide

Total : 0 €

DevCloud Shop | Td6 - Classroom

Not secure shop.devcloud.local/index.php

🛒 DevCloud Shop

Serveur : web2

Catalogue Produits

- Laptop - 800 € Ajouter
- Smartphone - 500 € Ajouter
- Headphones - 100 € Ajouter

🛒 Panier

- Headphones x 1 ❌
- Laptop x 1 ❌
- Smartphone x 1 ❌

Total : 1400 €

💳 Payer maintenant

DevCloud Shop | Td6 - Classroom

Not secure shop.devcloud.local/index.php

🛒 DevCloud Shop

Serveur : web2

✅ Paiement simulé avec succès !

Retour à la boutique

Conclusion

Ce TD avait pour objectif de déployer une architecture web haute disponibilité pour la startup DevCloud, composée d'un Load Balancer et de deux serveurs web., j'ai testé mon site dans plusieurs navigateurs.